Mechatronic Systems Laboratory

Project Report

WS 2024/25

Pick & Place Task Using Lego EV3 Manipulator

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Abstract

The Pick and Place operation involves a robot arm that retrieves an object from a designated location and places it at another location. This report focuses on a 2 degree of freedom manipulator robot performing an autonomous pick and place task. The task involves a homing operation initiated from a position described as home or start, which includes real-time distance measurement of objects using the ultrasonic sensor along with PID control to ensure precise joint angle control. The robot then secures the grip once it has captured an object and gently lifts the arm. The robotic arm employs inverse kinematics to navigate to the specified stations, showcasing its functionality in different scenarios. Achieving such progress necessitates the integration of PID control, inverse kinematics, sensors, encoders and optimized motor dynamics. This report outlines crucial steps for enhancing autonomy in both industrial and laboratory settings.

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1. Introduction

The robot utilized for this project is a Lego EV3 manipulator featuring 3 joints, 4 links, a gripper, and an assembly with 2 degrees of freedom (DOF). This manipulator can be manually controlled via buttons located on a control brick attached to its assembly. Alternatively, it can be programmed to execute specific tasks using either MATLAB or Simulink. Two motors have been implemented in the assembly at the initial two joints, whereas the third joint lacks a motor and acts as a constraint linking the components. An ultrasonic sensor is employed, positioned above the gripper and oriented vertically. It has kinematic constraints, which includes, link 1 being rigidly attached to link 2 at an obtuse angle of 135°, and Link 4 maintaining a perpendicular orientation to the ground plane. The rotation of link 3 is restricted by link 2 on the lower side and a touch sensor on the upper side. The base rotation is limited by a touch sensor on one side.

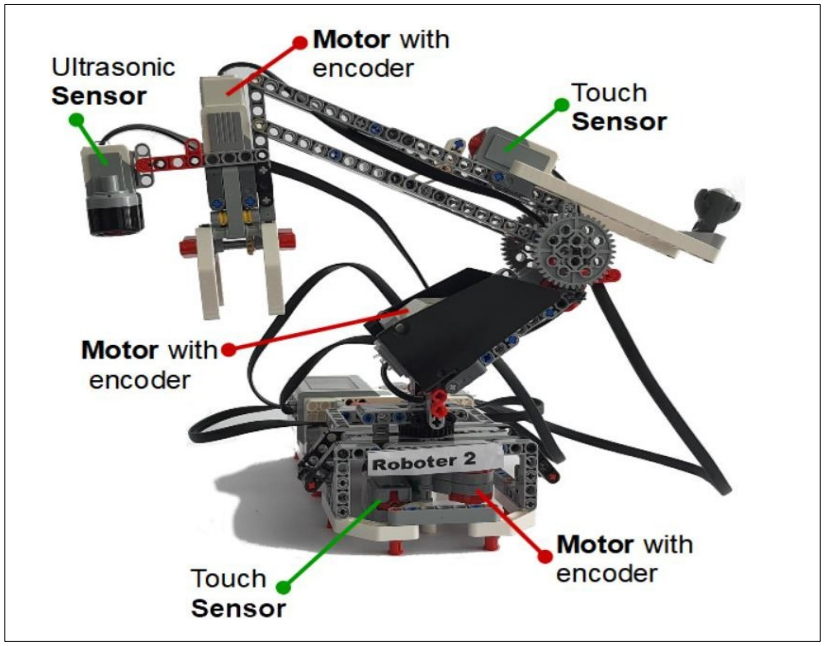

Fig. 1.1 Lego EV3 Manipulator

2. Components & Process flow chart

Each component is crucial for the manipulator's operation, aiding in accurate control and the execution of diverse tasks. The manipulator consists of three rotary actuating motors, each equipped with an encoder, along with an ultrasonic sensor and two touch sensors. Furthermore, all motors are connected via gearboxes to reduce load.

2.1. Servo Motors and Encoders

The manipulator as shown in the figure, is composed of three motors, all paired with encoders. The motor placed at the gripper manages the movement of the gripper, while the other motors at each respective joint facilitate the vertical movement of the gripper and rotation of the base, respectively, providing relevant feedback on orientation.

2.2. Touch Sensors

The manipulator is equipped with two touch sensors. The first touch sensor located at the base determines the range limits for the base motor, whereas the other touch sensor placed above the horizontal link establishes the limits for the joint motor, which enables accurate vertical movement.

2.3. Ultrasonic Sensor

The manipulator also utilizes an ultrasonic sensor, attached above the gripper to measure distances from objects. This sensor increases the manipulator's accuracy by supplying real-time distance measurements.

2.4. Gear Boxes

The gearboxes have been coupled with motors. The gearbox for joint motor, with a 1:5 reduction, provides greater torque for tasks that require significant force, while the gearbox for base motor, featuring a 1:3 reduction, is designed for applications requiring speed and accuracy.

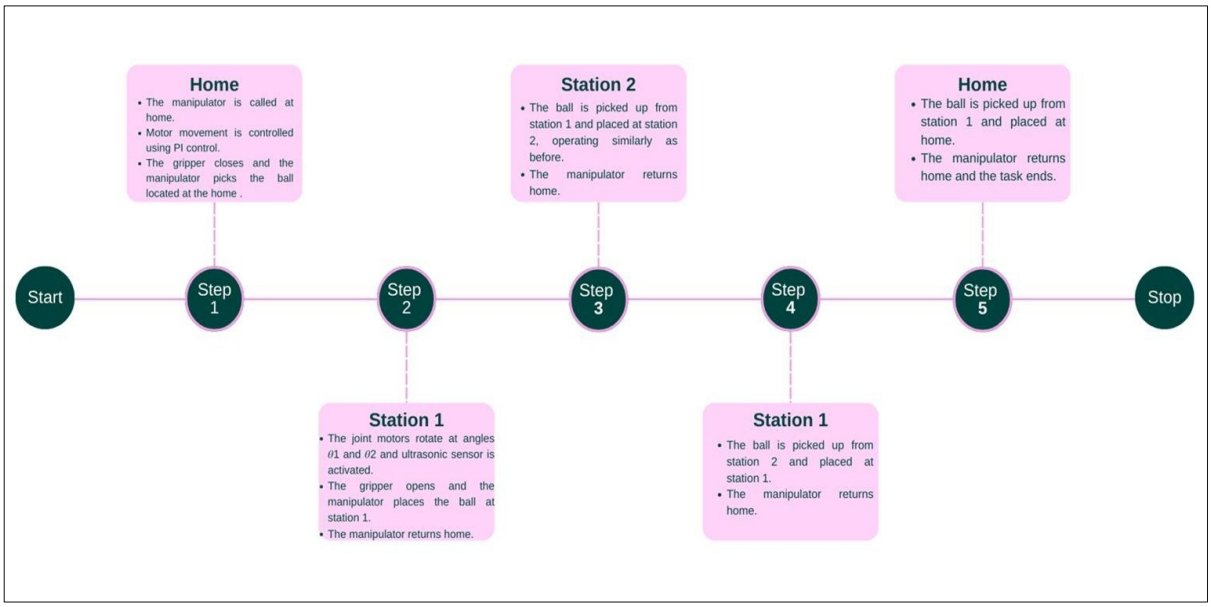

Fig. 2.1 Process flow chart

3. Inverse Kinematics

The motion of this 2-DOF manipulator robot (θ1 and θ2) can be controlled using two motors Motor B and Motor C whereas the gripper action can be controlled using Motor A. Assuming the Origin at the base of the robot (exactly below the Motor C) with X axis pointing outward, Y axis pointing to the right and Z-axis to the top when looked from the Front side of the robot. Using Graphical method and performing Inverse Kinematics, we can deduce the formulae for θ1 and θ2 from the given position of the gripper in rectangular co-ordinates X, Y, Z.

The inverse kinematics of manipulator robot is found by the following expressions:

L0 = 50mm (measured), L1 = 50mm, L2 = 95mm, L3 = 185mm, L4 = 110mm (already given)

$$\boxed{\Theta_1 = \tan^{-1}\left(\frac{y}{x}\right)}$$

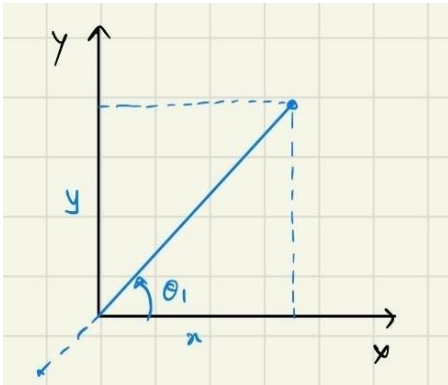

Fig. 3.1 Horizontal plane view of robot

$\sin(\Theta_2 - 45^o) = (L3 + P) / L3$

$P = Z - L0 - L1 - L2 * \sin 45^o$

$\Theta_2 = 45^o + [\sin^{-1}(Z - L0 - L1 - L2 * \sin 45^o / L3]$

$\Theta_2 = 45^o + [\sin^{-1}(Z - 57.175) / 185]$

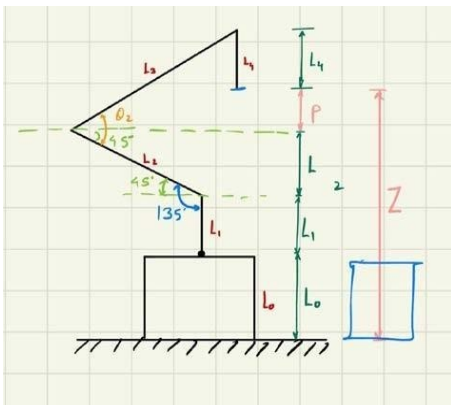

Fig. 3.2 Vertical plane view of robot

4. Program Implementation

Homing the manipulator: The manipulator's initial task is to align itself to the home position. This home position is specified as a location on the homing line, positioned somewhere beneath the gripper and ultrasonic sensor of the robot. When viewed from above, the homing line divides the manipulator into two equal parts, and it establishes the boundaries of the operating area. To perform the home operation, the manipulator is directed to the home position. The base motor rotates forward in small increments until the touch sensor is triggered, leading to a reset of the encoder. The joint motor then reverses until the touch sensor is activated, creating a reliable reference position for subsequent manipulations.

Pick and Place: The pick and place function involves the manipulator lifting a ball from designated locations, namely home, station 1, and station 2, and depositing it back at the same spots according to the pre-programmed sequence. The motors rotate at the specified angles and directions according to the pick and place operation between the various positions. Station 1 is situated 90 degrees to the left of the manipulator within the operating area, as seen from above. This position is the first to be included in the pick and place task. Station 2 is positioned at a 45-degree angle to the right concerning the homing line. This station contributes later in the process. The gripper operates by either opening or closing to manipulate the ball during this task. The ultrasonic sensor measures the distance from the gripper to the ball or ground, which triggers the gripper's opening or closing mechanism. Once the ball has been placed in the designated location, the manipulator is commanded to return to the home position to prepare for the next operation.

PI control: A combination of proportional and integral control has been implemented in the code to manage the regulation of the joint angle. The motor's initial rotation experiences a certain level of irregularity and inconsistency in rotational speed. To address this issue, the PI controller is designed to minimize the error, defined as the difference between the desired and actual angles, by adjusting the motor speed. These inputs respond not only to current errors but also to the accumulation of past errors. The controller's behaviour is influenced by parameters, specifically the proportional gain Kp (=0.1) and the integral gain Ki (=0.01). These parameter values were established by carefully considering the manipulator's movement for each case. As a result, the motor speed is governed by the interplay of these two factors.

5. Result

The system demonstrated a strong performance across all assignments. The inverse kinematic model facilitated precise control, while PI control enabled accurate tuning of the joint angles. The behaviours that were implemented allowed for smooth execution of homing, pick-and-place, and engagement with the distance sensor. In conclusion, the manipulator successfully accomplished complex tasks during the experiments.

6. References

[1] Mechatronic Systems Laboratory course contents WS 2024/25, https://moodle.uni-siegen.de/course/view.php?id=1438

[2] MATLAB Support Package for LEGO MINDSTORMS EV3 Hardware https://de.mathworks.com/help/supportpkg/legomindstormsev3io/